

Thinker AI

Complete Architecture & Brand Specification

Full Edition — June 2026

This is the single source of truth for Thinker AI — covering the complete V1 system architecture (Sections 1–27) and Brand Identity Guidelines (Section 28).

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1.

Purpose of This Document

The three founding constitutions describe Thinker AI's vision: a multi-agent system that understands goals, plans work, debates solutions, and delivers verified outcomes. This document translates that philosophy into a concrete, buildable V1 architecture defining every agent, data flows, technology choices, and implementation order.

2.

V1 Scope Statement

2.1 What V1 Is

A web-based chat application where a dynamic pipeline of thirteen lightweight agents — coordinated by Thinker Core — clarifies the request, thinks strategically, plans, builds a deliverable, reviews it, and returns one final answer. Supports 100+ languages automatically.

2.2 What V1 Is Not

- Not a fully autonomous multi-day project system.
- Not a mobile app — V1 ships as a responsive web app.
- Not running every request through full multi-model debate.
- Not retaining credentials, passwords, or payment data in memory.

2.3 V1 Success Criteria

#	Criterion
1	User can chat normally and get direct answers without triggering the full pipeline
2	User can request a project and the system asks clarifying questions before building
3	All thirteen agent roles are present in code
4	Every final deliverable passes through Reviewer before being shown to the user
5	Multi-model consensus works end-to-end for Judge Agent
6	Project memory persists within a session and reloads on return
7	Visual assets require explicit user approval before project is marked complete
8	Smart Clarification detects vague goals and activates 3-Level Deep mode
9	Signature Question fires on every project-type request before Builder runs

3.

Core Philosophy → Implementation Mapping

Constitution Principle	V1 Implementation
Understand first, plan second, execute third, verify always	Dynamic pipeline order enforced in code — routing adapts on failure
Requirement Clarity Law — ask minimum questions before large work	Clarification Agent hard gate — Builder cannot run until requirements are complete
Token Economy / Resource Protection Law	Cheap single-model calls by default; multi-model consensus only for high-stakes decisions
One Goal. One Team. One Final Answer.	Thinker Core merges all agent outputs into exactly one response object
No single model/agent/opinion is final — verified consensus is final	Judge Agent approval required before Thinker Core marks a deliverable final

The Founder Law

Thinker AI must not immediately build what the user asks for. It must first determine whether the user's goal, problem, and assumptions are valid. No code is ever written before the thinking is done.

The Preview Before Build Law

Thinker AI must show the user what it intends to build before generating the final deliverable. User approval is required at each stage before the next begins.

4.

System Architecture ★ REVISED

4.1 Pipeline Overview — Dynamic Routing

The pipeline no longer runs in a fixed sequence. Thinker Core operates as a dynamic routing engine — after every agent completes, it reads `next_action` and decides the next step.

Condition	next_action	Thinker Core Response
Simple chat / general question	direct_answer	Skip pipeline → single LLM call → return
Request is vague or incomplete	clarify	Halt → return questions to user
Requirements complete, project request	proceed	Run full pipeline in order
Builder output rejected by Reviewer	retry	Re-run Builder (same plan, same step)
Reviewer + Critic both flag deep issue	replan	Return to Planner with failure context
Planner→Builder loop exceeds 2 cycles	clarify	Return to Clarification for new requirements
Judge score borderline	escalate	Invoke Consensus Agent
All models for an agent unavailable	failover	Model Pool Manager assigns next provider
Any agent exceeds loop_count limit	halt	Pause pipeline, notify user, save state

4.2 Thinker Core Responsibilities ★ REVISED

- Owns the pipeline state machine and dynamic routing table for each conversation.
- Dynamic next-step decision: reads `next_action` from every agent output before passing control.
- Model Pool Manager: maintains a pool of available models per tier. If a model fails or times out (5s), silently reassigns to the next available model.
- Loop detection: tracks `loop_counts` per agent. Exceeding thresholds triggers halt and notification.
- Peer audit trigger: when a failover model takes over, instructs it to audit previous output first.
- State preservation on halt: writes full Pipeline State to database. Resume always possible.

4.3 Domain Picker

Named domain agents (Music, DevOps, Security, Coding, etc.) provide a domain hint to the pipeline — they are not alternate paths. Intent Agent is skipped when a domain is already selected. Closing the domain-picker→backend gap is Phase 10 in the build order.

5.

Agent Specifications ★ REVISED

All agents include a new output field read exclusively by Thinker Core: `next_action` (`proceed` | `retry` | `replan` | `clarify` | `escalate`) and `reason` (string). Agents recommend `next_action` — they do not execute it.

5.1 Intent Agent

Role	Classify the incoming message. Outputs domain label and Thinking Level. Skipped when user selects a domain.
Input	Raw user message + last 5 messages of context.
Output	Intent type, confidence, domain, <code>thinking_level</code> .
V1 Implementation	Single LLM call.
Model Tier	Fast/cheap.

5.2 Clarification Agent ★ UPDATED

Role	Enforce Requirement Clarity Law AND Smart Clarification — detect emotional context, surface hidden goals, challenge assumptions. Most critical agent: Wrong Goal → Wrong Strategy → Wrong Plan → Wrong Output.
Input	Intent type, full conversation, emotional tone signals.
Output	Completeness flag, smart questions, known requirements, assumption flags, Signature Question response.
V1 Implementation	LLM call with Smart Clarification Layer + 3-Level Deep + Constraint & Assumption Discovery + Signature Question Protocol.
Model Tier	Fast/cheap with enhanced prompt engineering.

5.3 Planner Agent

Role	Convert a clarified goal into an ordered list of concrete steps using the Strategy Agent brief.
Input	Known requirements + Strategy brief + Signature Question response + Constraint findings.
Output	Steps: <code>id</code> , <code>description</code> , <code>output_type</code> , <code>dependencies</code> .
V1 Implementation	Single LLM call.
Model Tier	Mid-tier reasoning.

5.4 Research Agent

Role	Gather external information when a step requires it. Optional — only invoked if Planner flags a step.
Input	Single Planner step flagged as needing research.
Output	Findings and sources.
V1 Implementation	Uses web search. Skipped for purely generative steps.
Model Tier	Fast/cheap with search.

5.5 Builder Agent

Role	Produce the actual deliverable — code, copy, config, or content. Most token-heavy agent.
Input	Step description, research findings, project memory.
Output	Artifact type, content, generated files.
V1 Implementation	Runs once per step, sequentially in V1.
Model Tier	Strongest available.

5.6 Reviewer Agent

Role	QC checklist: accuracy, logic, security, completeness, user-goal alignment. For images: relevance and appropriateness only.
Input	Builder output + original requirements.
Output	Pass/fail flag and list of issues with severity.
V1 Implementation	Single LLM pass. Fails with high-severity issues → returns to Builder once (capped at one retry).
Model Tier	Mid-tier reasoning.

5.7 Critic Agent

Role	Independently look for weaknesses Reviewer may miss. Prompt withholds Reviewer's verdict until Critic forms its own opinion.
Input	Builder output + Reviewer findings.
Output	List of concerns with overall severity.
V1 Implementation	Single LLM call with adversarial/skeptical framing — different persona from Reviewer.
Model Tier	Mid-tier reasoning.

5.8 Judge Agent

Role	Score on accuracy, feasibility, safety, cost, completeness. High-risk categories (auth, payment, data deletion, permissions) trigger Consensus automatically.
Input	Builder output, Reviewer result, Critic concerns.
Output	Score per criterion and approval decision.
V1 Implementation	If clearly high/low, Judge decides alone. If borderline, invokes Consensus.
Model Tier	Strongest available.

5.9 Consensus Agent

Role	Multi-model consensus for tie-breaking or mandatory high-risk categories. Three personas: Conservative Reviewer, Pragmatic Engineer, Critical Skeptic — never provider names.
Input	Same question Judge was uncertain about.
Output	Each model's vote and reasoning, plus final verdict.
V1 Implementation	Sends identical question to 3 different providers in parallel. Majority vote wins; ties → conservative outcome.
Model Tier	Three different providers.

5.10 Design Agent

Role	Visual/graphic output — logos, illustrations, UI imagery, icons. No 3D or animation.
Input	Planner step flagged output_type: image + style requirements.
Output	Generated image files + short description.
V1 Implementation	Single call to external image-generation API. Output passes through Reviewer.
Model Tier	External image provider.

5.12 Strategy Agent

Role	Business, product, and market thinking before technical planning. Runs after Clarification, before Planner, on project-type requests only.
Input	Clarified requirements + domain hint + Signature Question response + Constraint findings.
Output	Strategy brief: goal clarity, MVP scope, key risks, success criteria.
V1 Implementation	Single LLM call with structured prompt for business/product thinking.
Model Tier	Mid-tier reasoning.

5.13 Document Agent ★ NEW

Role	Read, parse, and extract structured content from any document file. 13th agent role.
Input	Uploaded file (PDF, DOCX, TXT, MD, CSV, XLSX) + user intent + context.
Output	Structured content: {summary, key_points, sections[], data_tables[], entities[], language, page_count}.
V1 Implementation	Two-step: (1) file parsing library extracts raw text; (2) single LLM call structures the output.
Model Tier	Fast/cheap.

5.2.1 Goal Discovery Layers

Layer	Purpose	Example Question
1. Goal Discovery	What does the user actually want to achieve?	"Why do you want to build this?"
2. Problem Discovery	What specific problem is being solved?	"What problem will this solve?"
3. Audience Discovery	Who is this for?	"Who will use this?"
4. MVP Discovery	Minimum viable starting point	"Which 3 features if you could only launch with 3?"
5. Success Discovery	What does success look like?	"How will you know this worked in 90 days?"

5.2.2 Smart Clarification Layer

- Emotional Context Detection: reads tone, not just words. Softens language when stress or urgency detected.
- Contradiction Detection: catches conflicts between later requests and earlier stated constraints.
- Assumption Surfacing: identifies unstated assumptions and confirms them before Planner locks scope.
- Progressive Disclosure: questions adapt based on previous answers — irrelevant follow-ups skipped.
- Intent Confidence Threshold: 90%+ → proceed; 70–90% → one confirmation; below 70% → Goal Discovery Mode.

5.2.5 Signature Question Protocol

"Why do you believe this is the right solution?" — asked on every project-type request before Builder is ever called. Hard rule: Clarification Agent never blocks a user who insists on proceeding — the user's autonomy is always respected.

6.

Model & Cost Strategy ★ REVISED

6.1 Model Tiers ★ REVISED

Tier	Used By	Purpose	Pool	Providers
Fast/cheap	Intent, Clarification, Research, Document	High-frequency, low-complexity	3	Min 2
Mid-tier	Planner, Strategy, Reviewer, Critic	Structured reasoning	3	Min 2
Strongest	Builder, Judge	Highest-stakes generation	3	Min 3
Image generation	Design Agent	Visual asset generation	2	Min 2
Multi-model	Consensus Agent	Tie-breaking votes	3	Exactly 3

Provider Diversity Rule: no two models in the same agent pool may share the same provider. A provider-wide outage removes at most one model from any pool.

6.4 When Consensus Actually Fires

Judge Score	Action
Clearly passes all five criteria	Approve directly — no Consensus call
Clearly fails (safety or accuracy critically low)	Reject — send back to Builder
Borderline on any criterion	Invoke Consensus Agent for that criterion only
High-risk category (auth, payment, data deletion, permissions)	Consensus fires automatically regardless of score

6.5 Model Identity Concealment

No provider or model name (Claude, GPT, Gemini, or any other) appears anywhere the user can see — not in chat, not in errors, not in debug views. Each model slot uses a generic label (Agent A, Agent B). Failures described generically: 'one of the verification steps didn't respond' — never naming the provider.

Data Flow & State ★ REVISED

7.1 New Pipeline State Fields

New Field	Type	Purpose
current_agent	string	Which agent is currently executing
routing_history	array	{agent, next_action, reason, model_used, timestamp} — full routing trace
loop_counts	object	{planner: int, builder: int, reviewer: int, ...} — per-agent retry counter
failover_log	array	{agent, failed_model, replacement_model, timestamp, peer_audit_result}
peer_audit_flag	boolean	Set true when failover model must audit previous output before continuing
resume_from_agent	string null	Set on halt — identifies restart point on resume
signature_question_response	string null	User's answer to Signature Question or null if declined
constraint_findings	object	Time, budget, skill, network, existing assets
assumption_flags	array	Surfaced assumptions and whether each was confirmed

7.2 Memory System

Memory Type	Scope	Storage	Retention
Short-term	Single conversation turn	In-memory / session	Cleared on session end
Project memory	One project across its lifecycle	Database, keyed by project	Until user deletes project
Long-term preference	User-level, across projects	Database, keyed by user	Indefinitely unless cleared
Decision Memory	Explicit saved decisions	Database, keyed by user	Until user revokes

Hard rule: passwords, PINs, API keys, and credentials are never written to any memory layer.

8.

Agent Fallibility & Error Containment ★ REVISED

8.2 Containment Rules

Rule	What It Prevents
No single agent's approval is final — Judge approval always required on top of Reviewer	A single fallible verdict reaching the user unchecked
If Reviewer and Critic disagree, Thinker Core routes to Consensus rather than letting Judge break the tie alone	A disagreement getting silently resolved by whichever agent runs last
Critic's prompt withholds Reviewer's verdict until Critic has formed its own opinion	Critic anchoring on 'Reviewer said it's fine' and skipping its own check
High-risk categories (auth, payment, data deletion, permissions): Consensus fires automatically	Shared blind spot across same-family models approving dangerous code
Every rejection or override logged with both verdicts and reasoning	Loss of visibility into how often agents disagree

8.4 Peer Recovery Protocol ★ NEW

Step	Action
1 — Failure Detection	Thinker Core detects failure when a model returns error, times out after 5s, or returns malformed output.
2 — Context Handoff	Thinker Core passes full Pipeline State to replacement model, including partial output and peer_audit_flag: true.
3 — Peer Audit	Replacement reviews what failed model produced — identifies what is correct, incorrect, and missing. Audit written to failover_log.
4 — Targeted Continuation	Replacement generates only what is needed — keeps correct partial output, fixes incorrect, completes missing. Never regenerates correct work.
5 — Normal Pipeline Resumes	Replacement model's output treated identically to a first-attempt output and proceeds normally.

8.5 Loop Detection Rules ★ NEW

Loop Type	Threshold	Response
Single agent retry	3 attempts	Halt. Save state. Notify user.
Planner → Builder → Reviewer cycle	2 full cycles	Return to Clarification with failure context. Ask user to re-scope.
Clarification loop (vague answers)	4 rounds	Surface the specific unknown directly.

9.

Pipeline Observability & Post-Delivery Feedback

9.2 Per-Agent Logging

Field	What It Records
agent_name	Which agent ran
input_summary	Short description of what this agent received
output_summary	Short description of what this agent produced
duration_ms	How long the agent took in milliseconds
status	success failed skipped
error_detail	If failed — error message using generic terms, never provider names
confidence	Judge's score, Consensus vote breakdown, Reviewer pass/fail
retry_count	How many times this agent was retried
failover_cost	Token spend from failover calls, separate from normal pipeline cost
clarification_depth	Which Clarification layers ran (Clarification Agent only)
signature_q_answered	Whether user answered the Signature Question

9.4 Post-Delivery Feedback Loop

User's Problem	System Response
Bug in the output	Ask user to describe bug → send back to Builder with bug + original plan → Reviewer re-runs after fix
Missing feature	Treat as new small requirement → Clarification → Planner (missing piece only) → Builder → Reviewer
Style not to taste	For images: Visual Asset revision flow. For code: ask for specific style feedback → Builder
Vague ('it's bad')	Ask one question: 'Is there a specific feature broken, something missing, or does it look wrong?'

9.5 Fix Limits — Cost-Based

Fix Type	What Runs	Limit
Small (color change, label update)	Builder only	Unlimited

10.

Technology Stack (V1 Recommendation)

Layer	Choice	Reasoning
Frontend	React + Vite	Matches existing developer experience
Backend	Node.js (Express) or Firebase Cloud Functions	Lightweight, matches existing Firebase familiarity
Database	Firestore	Document model fits the pipeline state object well
Auth	Firebase Auth	Reuse existing pattern
LLM Access	Direct API calls to each provider, backend only	Protects API keys, matches Resource Protection Law
Payments	Stripe	Webhook-based plan tier updates; never touches raw card data
Hosting	Firebase Hosting or Vercel	Simple, matches existing deployment habits

11.

Build Order ★ REVISED

Phase	Deliverable
1	Thinker Core skeleton + direct chat path (passthrough)
1.5 ★ NEW	Dynamic Routing Engine: routing table, next_action handling, loop_counts, loop detection, Model Pool Manager skeleton
1.6 ★ NEW	Model Pool Manager — failover detection (5s), pool health monitoring, Peer Recovery Protocol, failover_log
2	Intent Agent + Clarification Agent (basic) + hard gate
2.5	Smart Clarification Layer (Section 5.2.2)
2.6	3-Level Deep Clarification + Signature Question (Sections 5.2.3, 5.2.5)
2.7	Constraint Discovery + Assumption Discovery (Section 5.2.4)
3	Planner Agent + Builder Agent (single-step)
4	Reviewer Agent + one-retry loop back to Builder
5	Critic Agent + Judge Agent (Judge decides alone)
6	Consensus Agent + borderline-routing from Judge

V1 Launch Checklist

- Stable chat experience — direct path works without errors
- Full agent pipeline completes end-to-end on app, website, and game project types
- Strategy Agent produces a meaningful brief visibly reflected in Planner's output
- Decision Memory saves, applies automatically, and surfaces conflicts correctly
- Design Agent produces a usable image; approve/reject/revise all work; 3-attempt cap enforced
- Clarification gate cannot be bypassed
- Smart Clarification detects emotional tone and adjusts question framing
- Contradiction Detection catches conflicts between later messages and earlier constraints
- Signature Question fires on every project-type request; response stored in Pipeline State
- Intent Confidence Threshold routes: high → proceed, mid → confirm, low → Goal Discovery
- No credentials ever appear in stored memory (manual audit)
- Consensus Agent tested with intentionally borderline cases
- High-risk categories (auth, payment, data deletion, permissions) trigger Consensus automatically
- Token cost per request is logged and reviewed
- Final response is always a single merged answer — never raw agent dumps
- 5-stage output pipeline works end-to-end (Section 16)
- Language detection works — user writes in any Tier 1 language, all responses in that language
- RTL language (Arabic or Hebrew) renders correctly
- Per-agent log entries written for every pipeline run; failed run traceable to specific agent
- Cost-based fix limits enforced: small unlimited, medium capped at 10, full rebuild capped at 3
- Version history saves correctly; rollback restores without Builder call; 10-version limit enforced
- Dynamic routing engine tested — all next_action values handled correctly
- Failover tested — model timeout triggers pool reassignment within 5 seconds
- Peer Recovery Protocol tested — replacement model audits and continues partial work correctly
- Loop detection tested — exceeding thresholds triggers halt and state save
- Think Credits deducted correctly; no silent overruns; confirmation shown above 3 credits
- Stripe webhook correctly updates plan tier on payment and downgrade
- File upload security rules tested — executables, oversized files, credentials all blocked
- Document Agent correctly structures output from PDF, DOCX, and TXT uploads
- No provider/model names reachable from any user-facing surface including error states

13.

Explicitly Out of Scope for V1

- Parallel multi-step Builder execution
- Automatic re-planning when a step fails repeatedly
- Native mobile app (Play Store launch can wrap the web app initially)
- Organisational/team-shared memory
- Autonomous multi-day project system with no human checkpoint
- Native Thinker foundation model (V1 uses third-party model APIs only)
- 3D model generation, animation, and video generation
- Interactive prototypes or Figma-format design file output
- Live API testing in preview (V2 feature)

14.

Open Decisions for Founder Review

1. Consensus uses exactly three providers — confirmed for V1, configurable in later versions.
2. Reviewer retry capped at one loop — confirm this is acceptable or should be configurable.
3. Research Agent is optional and skippable per step — confirm this matches intended behaviour.
4. Technology stack assumes continuity with existing Firebase project — confirm versus fresh stack.
5. High-risk category list (auth, payments, data deletion, permissions) — confirm completeness.
6. Design Agent image provider not yet chosen (DALL-E vs Stability AI) — confirm based on cost and licensing.
7. Free Trial credit amount set at 50 (one-time) — confirm conversion funnel balance.
8. Pro price at \$19/month, Founder at \$59/month — confirm based on competitive positioning.
9. Critic Agent skipped on Free plan to reduce cost — confirm acceptable quality trade-off.

15.

Goal Intelligence System

15.1 Goal Discovery

Works through five layers of questions — one at a time, conversationally — to surface the real goal beneath the stated one. Goal Discovery is not an obstacle — it is the product.

15.3 Idea Validation

Strategy Agent runs Idea Validation covering: (1) Is the problem real? (2) Is this idea the right solution? (3) Is this the right time? (4) Why this team? Output is a brief that sharpens the plan — not a go/no-go block.

15.5 Founder Mode

Normal Mode	Founder Mode
Goal clarity for this build	Problem → Market → Competition → Business model
MVP scope (features)	MVP scope (minimum to validate the business)
Key technical risks	Full risk landscape: market, technical, financial, execution
Success criteria for deliverable	Success criteria for the business itself

16.

Output Pipeline — Preview Before Build

Sta ge	Name	User Sees	User Action
1	Thinking Summary	Goal understanding, strategy brief, proposed approach	Confirm or correct
2	Blueprint Preview	Full plan: screens/components, tech stack, estimated complexity	Approve, edit, or request changes
3	Build Preview	Real-time progress + live rendered UI. Mandatory approval after first screen.	Approve first screen or change direction
4	Approval	Full build visible in preview panel	Approve, change something, or go back to blueprint
5	Final Output	Project summary, Live Preview link, ZIP download, Architecture notes	Download and use

17.

Multi-Language Support

Content Type		Language	Reason
All conversation with user		User's language	Product interface
Clarification questions, strategy brief		User's language	Must be clearly understood
Progress indicators, error messages		User's language	UX must be native-feeling
Generated code (variables, functions)		English	Industry standard
Code comments		User's language or English (user's choice)	Via Decision Memory
README and documentation		User's language	User needs to understand their own project
Folder/file names, API endpoint names		English	Compatibility and portability
Tier	Languages		Quality
Tier 1 — Full	English, French, Spanish, German, Portuguese, Italian, Dutch, Japanese, Korean, Chinese, Arabic, Hindi, Bengali, Turkish, Russian, Polish, Vietnamese, Indonesian		High quality across all agents
Tier 2 — Good	50+ additional including Urdu, Malay, Thai, Swahili, Romanian, Czech, Swedish, Greek, Hebrew, and others		Good quality; occasional phrasing issues
Tier 3 — Basic	Remaining languages supported by underlying model providers		Varies; user informed if quality may be limited

18.

Paid Plan Architecture

Tier	Price	Think Credits	Target User
Free Trial	\$0	50 (one-time, never refills)	Individuals exploring the product
Pro	\$19/mo or \$199/yr	1,500/month	Freelancers, indie builders
Founder	\$59/mo or \$589/yr	5,000/month	Power users, startup founders
Enterprise	Custom	Custom	Organisations 10+ seats (V2)

18.2 Feature Gate Matrix

Feature	Free Trial	Pro	Founder
Direct Chat	✓	✓	✓
Basic Clarification	✓	✓	✓
Smart Clarification (3-Level, Sig Q)	—	✓	✓
Strategy Agent	—	✓	✓
Founder Mode	—	✓	✓
Planner Agent	✓	✓	✓
Research Agent	—	Limited	Unlimited
Builder Agent	Credit-limited	Credit-limited	Credit-limited
Reviewer + Judge	✓	✓	✓
Critic Agent	—	✓	✓
Consensus Agent	—	—	✓
Design Agent	—	✓	✓
Document Agent	✓	✓	✓
5-Stage Output Pipeline	Stage 1+5 only	Full	Full
Version History	3 versions	10 versions	25 versions
Decision Memory	—	—	✓
Thinking Level: High (■)	—	✓	✓

Feature	Free Trial	Pro	Founder
Thinking Level: Consensus (■)	—	—	✓
Priority Queue	—	—	✓
API Access	—	—	✓
Extra credit rate	\$0.10/credit	\$0.07/credit	\$0.05/credit

18.3 Billing Integration

- Stripe Checkout handles all payment. Thinker AI never touches raw card data.
- On successful payment, Stripe webhook updates plan_tier in Firestore and refreshes session token.
- Downgrade takes effect at end of current billing period. Pro-created projects preserved but gated features become read-only.
- Failed payment: 7-day grace period. After 7 days without resolution, tier reverts to Free Trial.
- Refunds handled entirely in Stripe — plan_tier change is the only signal Thinker Core needs.

19.

Resilience — Continuous Delivery Guarantee

Thinker AI never holds a user's work hostage to a single model, a single provider, or a temporary outage. When any model fails, another model takes the task, audits what was done, and continues. The user's work always moves forward.

Time	Event	Thinker Core Action
0s	Model call initiated	Start 5-second timeout clock
0–5s	Model responds normally	Continue pipeline — no action needed
5s	No response received	Mark model as failed. Assign next model from pool.
5–7s	Replacement model initialises	Peer audit begins (Section 8.4 Steps 2–3)
7s+	Replacement model continues task	Pipeline resumes — user sees no pause
All pool models fail	Pool exhausted	Loop detection triggers halt. State saved.
Situation		Message to User
Failover resolves within 7s		No message. Pipeline continues normally.
Failover takes 7–30s		'Still working on this — taking a moment longer than usual.'
Partial completion, halt		'I've completed [X of Y steps]. Your work is saved. Would you like me to try a different approach?'
All models unavailable		'I wasn't able to complete this step right now. Your work is saved and you can resume at any time.'

Dynamic Routing Engine

Routing Decision Logic (pseudocode)

```
function routeAfterAgent(agent_output, pipeline_state):
    action = agent_output.next_action
    log_routing_decision(agent_output, pipeline_state)
    if action == 'proceed':
        return dispatch(next_in_sequence(pipeline_state.current_agent), pipeline_state)
    if action == 'retry':
        pipeline_state.loop_counts[current_agent] += 1
        if loop_counts[current_agent] > 3: return trigger_halt(pipeline_state)
        return dispatch(current_agent, pipeline_state)
    if action == 'replan':
        pipeline_state.loop_counts['planner_cycle'] += 1
        if loop_counts['planner_cycle'] > 2: return dispatch('clarification', pipeline_state)
        return dispatch('planner', pipeline_state_with_failure_context)
    if action == 'clarify': return dispatch('clarification', pipeline_state)
    if action == 'escalate': return dispatch('consensus', pipeline_state)
    if action == 'direct_answer': return dispatch('direct_llm_call', pipeline_state)

function callAgentWithFailover(agent_name, input, pipeline_state):
    pool = get_pool(agent_name)
    for model in pool.available_models:
        try:
            result = call_model(model, input, timeout=5s)
            if is_valid_output(result): return result
        except TimeoutError, ModelError:
            log_failover(agent_name, model, pipeline_state)
    next_model = pool.next_available(exclude=failed_models)
    if not next_model: return trigger_halt(pipeline_state)
```

21.

Think Credits, Pricing & Thinking Levels

More Thinking
= More Questions
= More Validation
= More Credits

Users pay for the depth of thinking — not just the answer. Every interaction invokes a third-party model API call, so every interaction costs at least 1 credit.

Level	Icon	Purpose	Agents Active	Base Cost
Low	■	Fast answers, simple chat	Intent + Direct Answer	1 credit
Medium	■	Analysis, comparisons	Intent + Clarification + Planner + Builder	3 credits
High	■	Deep planning, full project builds	Full pipeline (all 13 agents)	10 credits
Consensus	■	Multi-model validation	Full pipeline + Consensus (3 providers)	30 credits
Thinking Level		Free Trial	Pro	Founder
■ Low		✓	✓	✓
■ Medium		✓	✓	✓
■ High		—	✓	✓
■ Consensus		—	—	✓

Credit Cost Table

Action	Credits	Notes
Simple chat / direct answer	1	Per message. Always applies.
Clarification reply (per round)	2	Each round of questions + user answer
Research Agent (per step)	3	Only when Planner flags a step as research-needed
Medium analysis	4	Medium Thinking sessions
Image generation (per image)	10	Per image, per attempt including revision
High Thinking session (base)	10	Charged once at session start
Consensus session (base)	30	Charged once at session start

22.

File Upload Architecture

Thinker AI accepts any file. Thinker Core identifies the type, routes to the correct agent, and determines what to do — the user does not need to specify what to do with the file.

Category	Formats	Max Size	Routed To	Credits
Documents	PDF, DOCX, TXT, MD	50 MB	Document Agent	3/file
Code	JS, TS, PY, HTML, CSS, JSON, YAML, all code	10 MB	Reviewer → Critic → Judge → Builder	5 review / 10 fix
Images	PNG, JPG, JPEG, SVG, WEBP, GIF	20 MB	Design Agent	2 analyse / 15 edit / 20 ref
Data	CSV, XLSX, JSON	25 MB	Research Agent	3/file
Archives	ZIP, TAR	100 MB	Thinker Core (extract & re-route each file)	5 + per-file costs

Image Operations

Operation	User Intent Signal	Design Agent Action	Credits
Analyse only	No edit instruction	Vision model describes content, objects, colours, style	2
Edit / remove object	'Remove background' / 'Delete the person on the left'	Image inpainting / masking API	15
Style transfer	'Make this cartoon style' / 'Oil painting effect'	Style transfer API applied to full image	15
Reference-based generation	'Use this logo's style to make a banner'	Style extracted; new asset generated using extracted style	20
Upscale / enhance	'Make this higher resolution' / 'Sharpen this'	Upscaling API applied	8

File Security Rules

- Executable files (.exe, .bat, .sh, .dll, .bin, .dmg) are rejected before processing — no credits charged.
- No file stored permanently — deleted when session ends.
- File content sent to LLM providers as context only — never uploaded to separate file storage.
- Code files scanned for: shell injection, data exfiltration, cryptomining, known malware signatures.
- If a code file contains an API key or password, Thinker Core warns the user and redacts the value from LLM context.

UI/UX Specification

Zone	Location	Contents
Left Sidebar	Fixed, collapsible	Logo, Domain Picker, Conversation History, New Chat, Upgrade CTA
Main Chat Area	Centre, scrollable	Message thread, Thinking Level indicator, pipeline stages, output panels
Input Bar	Bottom, fixed	Text input, file upload, Thinking Level selector, send button, credit balance chip
Right Panel	Slide-in on demand	Project details, version history, pipeline log (Founder), settings shortcuts
Top Bar	Fixed, minimal	Conversation title (editable), domain label, user avatar, notifications

Screen Layouts

Screen	Key Elements
Home	Greeting ('Good Evening') · 'What are we building today?' prompt · Bottom nav: Chat · Projects · Create · Files · Profile
Chat	Message thread · Thinking indicator between user msg and Thinker reply · Pipeline stage labels · Bottom input bar
Agent View	Hidden by default, expandable · Shows live pipeline stages with checkmarks and credit cost per stage
Project	Project name + status badge · Timeline: Thinking → Planning → Building → Testing → Completed · Download ZIP · Version history
Voice	Full-screen dark · Logo centered with Core Pulse animation · Minimal wave animation · Tap anywhere to stop
Settings	8 tabs: General · Thinking & Pipeline · Memory · Files · Credits & Billing · Notifications · Privacy & Security · Advanced (Founder)
Profile	Identity (name, email, photo) · Activity stats · Decision Memory manager · Linked accounts

Settings — 8 Tabs

Tab	Key Settings
General	App language · Response language · Default Thinking Level · Default domain · Theme · Font size
Thinking & Pipeline	Auto-select Thinking Level · Show credit cost preview · Show pipeline stages · Signature Question on/off · Goal Discovery depth
Memory	Project / Long-term / Decision Memory on/off · Clear all memory · Export all memory as JSON

Tab	Key Settings
Files & Uploads	Default action on code/document/image/ZIP upload · Auto-scan code for security · Warn before sending file to LLM
Credits & Billing	Current plan · Credits balance + breakdown · Buy extra credits · Upgrade/downgrade · Billing history · Auto top-up
Notifications	Pipeline complete · Build failed · Low credit balance · New features · Weekly summary · Billing (always on)
Privacy & Security	Change password · Two-factor auth · Active sessions · Data export · Delete account · Analytics opt-out
Advanced (Founder)	API key management · Webhook URL · Pipeline log · Pool health dashboard · Credit usage API

24.

API & Webhook Specification

Available to Founder plan only. Base URL: <https://api.thinker.ai/v1> · Auth: Authorization: Bearer tk_live_xxxx · Rate limit: 60 req/min per API key.

Method	Endpoint	Purpose
POST	/pipeline/run	Run the full pipeline on a prompt. Returns pipeline_id immediately (async).
GET	/pipeline/{id}	Get status and result of a pipeline run.
POST	/pipeline/{id}/respond	Send user response to clarification/approval prompt mid-pipeline.
DELETE	/pipeline/{id}	Cancel a running pipeline.
GET	/credits/balance	Get current Think Credit balance and monthly usage.
GET	/credits/history	Paginated list of credit transactions.
GET	/projects	List all projects for this account.
GET	/projects/{id}/versions	List all saved versions of a project.
POST	/files/upload	Upload a file for use in a pipeline run.

Webhook Events

Event	When Fired
pipeline.started	Pipeline begins running
pipeline.clarification_needed	Clarification Agent needs user input
pipeline.stage_complete	Each pipeline stage completes
pipeline.approval_needed	Stage 4 approval required
pipeline.complete	Pipeline finishes successfully
pipeline.failed	Pipeline fails unrecoverably
pipeline.halted	Loop detection or pool exhaustion halts pipeline
credits.low_balance	Balance drops below threshold
credits.depleted	Balance reaches 0
failover.occurred	Model failover happened (informational)

Webhook security: every request includes a Thinker-Signature header (HMAC-SHA256). Retries failed deliveries up to 5 times with exponential backoff (1m, 5m, 30m, 2h, 8h). After 5 failures, endpoint is disabled and founder is notified.

Testing & QA Strategy

Layer	What Is Tested	When Run
Unit tests	Thinker Core routing logic, credit calculations, loop detection — not LLM calls	Every commit (CI)
Agent mock tests	Each agent's input→output contract using deterministic mocks	Every commit (CI)
Integration tests (real models)	10 fixed test prompts against real models, scored by Judge Agent. Pass if score ≥70%.	Every staging deploy
Failover tests	Primary model blocked — pipeline must complete via secondary within 10 seconds	Weekly automated
Credit accuracy tests	Each action type credited exactly per Section 21 table	Every staging deploy
Security tests	Upload test files: malware pattern, executable, oversized, credential-containing code. All must be rejected.	Every staging deploy
Manual QA scenarios	End-to-end user flows that cannot be automated	Before each release

Key Manual QA Scenarios

Scenario	Key Assertions
A — Simple chat	■ Low auto-selected · 1 credit deducted · No clarification asked · Answer in <5s
B — Project build	Signature Question asked · Goal Discovery runs · All 5 stages visible · ZIP downloadable · Credits match estimate
C — Vague request	Intent confidence below 70% → Goal Discovery Mode · Questions one at a time · No Planner until discovery complete
D — Failover invisible	Primary model blocked · Build completes · No error shown to user · failover_log has one entry
E — Credit gate	Balance 3 credits · High Thinking request (10 credits) → confirmation modal · 'Insufficient' if confirmed with no top-up
F — Code review upload	File received card shown · Auto-review (5 credits) · Bug identified · Fix offered · Fixed file downloadable
G — RTL language	Write in Arabic · All responses in Arabic · UI right-aligned · Code stays in English · README in Arabic
H — Loop detection	Force Reviewer to always return retry · Builder retried exactly 3 times · Halt on 4th attempt · State saved

26.

Error Codes & Messages

Every error has a machine-readable code, a user-facing message (non-technical, in the user's language), and a developer-facing message. No provider or model name ever appears in any error message.

Code	User Message	Developer Message
AUTH_001	Please sign in to continue.	Missing or invalid Authorization header.
AUTH_002	Your session has expired. Please sign in again.	JWT token expired.
AUTH_004	This API key is invalid or has been revoked.	API key not found or revoked in database.
CREDIT_001	You don't have enough Think Credits. You have {balance} — this needs {required}.	Credit balance insufficient.
CREDIT_002	Your Think Credits have run out. Top up to continue.	Credit balance is 0.
PLAN_001	Strategy Agent is available on Pro and above.	Feature strategy_agent not available on plan: free_trial.
PLAN_002	Consensus Thinking is available on the Founder plan.	Feature consensus_thinking not available on plan: pro.
PLAN_003	You've reached your monthly project limit. Upgrade to Pro.	monthly_project_count >= plan_limit.
PIPELINE_001	I need a bit more information before I can start.	Pipeline rejected: requirements_complete=false.
PIPELINE_002	Something went wrong. Your work up to this point is saved.	Unhandled exception in pipeline at stage={stage_name}.
PIPELINE_003	I ran into a difficulty on this step. Your work is saved.	All models in pool exhausted for agent={agent_name}.
PIPELINE_004	This step has been retried too many times. Let's try describing requirements differently.	loop_count exceeded threshold.
AGENT_001	Still working on this — taking a moment longer than usual.	Model timeout after 5s. Failover initiated.
FILE_001	That file type isn't supported.	Unsupported MIME type: {mime_type}.
FILE_002	That file is too large. Max size is {limit}MB.	File size {size} exceeds limit for type {category}.
FILE_003	This file type can't be uploaded for security reasons.	Blocked file extension: {ext}.

Code	User Message	Developer Message
FILE_004	This code file appears to contain sensitive data. Please remove it before uploading.	Credential pattern detected in file.
FILE_005	This code contains patterns that look potentially harmful. It won't be built.	Malicious code pattern detected: {category}.
API_001	API rate limit reached. Please wait before making more requests.	Rate limit exceeded: {count} requests in 60s.
WEBHOOK_001	N/A	Webhook delivery failed after 5 attempts. Endpoint disabled, founder notified.

27.

Analytics & Business Metrics

Metric	V1 Target	Warning Threshold
Pipeline completion rate	≥ 70%	< 50% = reliability problem
Free → Pro conversion (30 days)	≥ 8%	< 3% = value not clear
Pro → Founder upgrade (60 days)	≥ 5%	< 2% = Founder plan differentiation unclear
Monthly churn (Pro)	≤ 5%	> 10% = product not delivering value
Avg credits / High Thinking build	40–80 credits	> 120 = cost model unsustainable
Operator LLM cost / High Thinking build	≤ \$0.60	> \$1.20 = margin unsustainable
Failover rate	≤ 5%	> 15% = provider reliability problem
Clarification abandonment rate	≤ 20%	> 35% = too many or wrong questions
Upgrade prompt CTR	≥ 12%	< 5% = prompts not persuasive
Support ticket rate	≤ 2 per 100 users/week	> 8 = UX or reliability problem

V1 Success Definition (60 days after launch): pipeline completion ≥70%, Free→Pro conversion ≥8%, Pro churn ≤5%, operator cost per High Thinking build ≤\$0.60, failover rate ≤5%, and positive qualitative user feedback visible.

Brand Identity Guidelines

Trait	What It Means in Practice
Intelligent	Reasons, validates, and challenges assumptions — never guesses
Trustworthy	Every output reviewed by multiple agents before the user sees it
Minimal	No noise, no clutter — every element serves a purpose
Premium	The experience feels crafted, not generated
Futuristic	Ahead of the curve in product, design, and thinking
Calm	Never urgent, never anxious — steady, focused intelligence

Taglines

Rating	Tagline	Best For
★★★★★ Primary	Think Beyond Intelligence	App, splash screen, primary marketing
★★★★ Secondary	Many Minds. One Intelligence.	Architecture, investor pitch
★★★★ Secondary	The AI That Thinks Before It Builds.	Product description, onboarding

Color System

Name	Hex	Use	Do Not Use For
Deep Teal	#0B6E69	Primary brand — buttons, links, active states, logo	Body text, large backgrounds
Teal Light	#14B8A6	Core glow, gradient end, accent highlights, code text	Backgrounds, paragraph text
Dark BG	#0B1220	App background, splash screen, full-bleed sections	Cards, modals
Surface	#111827	Cards, sidebars, modals, input fields	Main background
Divider	#1F2937	Dividers, card outlines, input borders	Text, icons
Text White	#F9FAFB	Headings, labels, important body text	Disabled states
Text Mid	#94A3B8	Body text, descriptions, secondary content	Headings
Gradient	#0B6E69→#14B8A6	Primary CTA buttons, progress bars	Backgrounds, body text

Typography

Font	Role	Sizes	Weight
Space Grotesk	All headings, navigation, brand name	Display 48 · H1 32 · H2 24 · H3 20 · H4 16px	Bold 700 / SemiBold 600
Inter	All body text, descriptions, captions	Body 16 · Body 14 · Caption 12 · Label 11px	Regular 400 / Medium 500
JetBrains Mono	All code blocks, API output, keys	13px	Regular 400

Logo Meaning

Element	Meaning
Center Core	Thinker Core — the orchestration brain of the system
Outer Nodes (×6)	The 6 primary AI Agent roles working in parallel
Connection Lines	Agent collaboration and data flow between all agents
Core Color (#0B6E69)	Active, focused intelligence — the system is thinking
Whole Symbol	Unified Intelligence — many minds, one trusted outcome

UI Design Rules

Property	Value
Default theme	Dark Mode (Light mode available but secondary)
Button radius	16px
Card radius	20px · 1px border · No shadows
Shadow	None — flat design system throughout
Transition	200–250ms ease-out — slow, smooth, no bounce
Icons	Lucide Icons — outline only — 1.5px stroke — 24×24px base
Loading	3 dots: ● → ●● → ●●● at 400ms intervals — never a spinner
Sound	3 soft sounds only: Thinking · Complete · Notification — off by default

Brand Voice

■ Never Say	■ Thinker AI Says
Here is your answer.	I analyzed your goal. Here's what I found.

■ Never Say	■ Thinker AI Says
Great question! I'd be happy to help!	Let me think through this.
Processing your request...	Analyzing your goal...
Done! Your project is ready.	Build complete. Reviewed by 4 agents. Here's what was built and why.
I might be wrong.	This is my best assessment — review before deploying to production.

App Open Animation — 2–3 Seconds

Time	What Happens
0ms	Black screen.
0–300ms	6 outer nodes fade in, 50ms apart, clockwise
300–600ms	Connection lines draw from nodes toward center
600–1000ms	Center Core brightens — Deep Teal glow pulses outward once
1000–1500ms	'Thinker AI' fades in letter by letter below logo
1500–2000ms	Tagline fades in: 'Think Beyond Intelligence'
2000–2500ms	Brief hold. Fade to Home Screen.

Do's & Don'ts

■ Always Do	■ Never Do
Use #0B1220 as the app background	Never use logo on white or light background
Use Space Grotesk Bold for all headings	Never add drop shadows to any element
Use Inter Regular for all body text	Never use 3D, neon, or rainbow effects
Use Lucide outline icons at 1.5px stroke	Never use filled icons — outline only
Show thinking progress — all pipeline stages	Never put text inside the logo symbol
Use gradient only on primary CTA buttons	Never say 'Great question!' or 'Happy to help'
Use 3 dots (●●●) for loading — never a spinner	Never show provider names (Claude, GPT, etc.)
Keep Core glow on Center Core only	Never use the gradient as a page background
Animate at 200–250ms — no bounce	Never use a spinner for loading
Keep sound off by default	Never bounce or spring any animation

Think Beyond Intelligence

Thinker AI Complete Specification — Sections 1–28 · June 2026